Jyot Shah

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PROFILE

Computer Science undergraduate specializing in AI systems, Machine Learning, and backend software engineering. Experienced in building full-stack AI applications, RAG pipelines, computer vision models, and scalable web platforms using modern LLMs and cloud services. Strong foundation in Data Structures and Algorithms, Object Oriented Programming, Operating Systems, and Database Systems, with a focus on solving real-world engineering problems.

EDUCATION

Manipal University Jaipur — 2023 – 2027
B.Tech. in *Computer Science & Engineering (AI & ML)* — **CGPA: 9.21**

WORK EXPERIENCE

AI Backend Developer | *Southern Federal University, Russia × Manipal University Jaipur, India*
(International Interdisciplinary Student Project) — Nov 2025 – May 2026

- Built the backend for *AssistMe*, a product intelligence platform using FastAPI, PostgreSQL, Grok LLM, and JWT authentication. Designed secure REST APIs, modular AI services, and automated product data ingestion pipelines to process 10K+ product specifications, enabling conversational recommendations and intelligent device comparisons.

ML Research Engineer (Code) | *Freelance* — Jan 2026 – May 2026

- Implemented and enhanced an IEEE Access ANN-based image encryption framework in Python, improving diffusion, key sensitivity, and nonlinear substitution through optimized XOR chaining, key-derived ANN mappings, and algorithmic refinements. Developed an automated benchmarking pipeline, achieving 7.9977-bit entropy, 99.67% NPCR, 33.57% UACI, and pixel-perfect decryption (SSIM = 1.0) while validating performance against multiple baseline architectures.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++
Full-Stack: React.js, HTML, CSS, EJS, Node.js, Express.js, Flask, RESTful APIs, MongoDB, MySQL, FastAPI
ML Frameworks: TensorFlow, Scikit-learn, Keras, OpenCV, Pillow, NumPy, Pandas
AI Systems: RAG Pipelines, Vector Embeddings, LLM Integration, Retrieval Systems, Prompt Engineering
Tools & Platforms: Git, GitHub, Vercel, Render, Google Colab, Kaggle Code, Redis, Qdrant

PROJECTS

Forge - AI Developer Workspace & Persistent Knowledge Engine (Code) (Live) | *MERN, Qdrant, Redis* — 2026

- Built an AI-powered developer workspace using a hybrid RAG pipeline with Google Gemini, MongoDB, Qdrant, and vector embeddings, enabling context-aware assistance across thousands of indexed documentation and code chunks.
- Engineered a backend with Redis, MongoDB GridFS, JWT authentication, and Reciprocal Rank Fusion (RRF), automating document ingestion, embedding generation, and asynchronous processing while reducing response latency by ∼40%.

Plant Leaf Disease Detection & Treatment System (Code) | *YOLOv11, Flask, Gemini* — 2025

- Built an end-to-end plant disease detection system using a custom-trained YOLOv11 model on an augmented PlantDoc dataset, achieving ∼96% validation accuracy with optimized preprocessing and real-time inference deployed via Flask.
- Integrated a Gemini 2.5 Flash based treatment chatbot and a responsive frontend, enabling instant annotated predictions and context-aware treatment recommendations, improving diagnostic interaction speed by ∼40%.

InfraGhost AI - Infrastructure Reality Verification System (Code) (Live) | *MERN Stack, Gemini Vision AI* — 2026

- Built a Gemini Vision powered full-stack platform for geotagged infrastructure analysis and 0–100 Ghost Score based asset classification; client-side image compression reduced payload size by ∼60%, improving AI response latency by ∼35%.
- Developed a hardened Express.js backend with rate limiting and input validation, integrated MongoDB storage, and designed a real-time Mapbox analytics dashboard with statistics and PDF reporting for scalable authority-level monitoring.

RESEARCH EXPERIENCE

Hierarchical Vision Transformer Ensemble for Robust Plant Leaf Disease Classification — Ongoing

- Curating and training on diverse plant leaf datasets (80K+ images, 30+ classes), using class-balanced sampling and controlled augmentations to mitigate domain shift; achieving ∼97% validation accuracy with improved dataset stability.
- Building a hierarchical ViT with parent–child attention, combined with lightweight ensembling and a hybrid loss (CE + Focal + adaptive margin), yielding 4–6% F1 gains over baseline CNN/ViT models while remaining GPU efficient.

ACHIEVEMENTS

Honoured with the Dean's List award for academic excellence in the 1st, 2nd, and 4th semesters, consistently ranking among the top-performing students in the Department of AIML at Manipal University Jaipur.